

CREATE VPC MANUALLY USING GCP CONSOLE

Step 1: Login to GCP Console

1. Go to <https://console.cloud.google.com>
2. Select the correct project (labs-487521)

Step 2: Navigate to VPC

■ Navigation Menu → VPC Network → VPC Networks

Step 3: Click "Create VPC Network"

Step 4: Configure VPC

Name: my-custom-vpc Subnet creation mode: Custom

Best Practice: ✓ Always use Custom mode for production ✓ Avoid Auto mode for enterprise environments

Step 5: Add Subnet

Subnet Name: subnet-us-central1 Region: us-central1 IP Range: 10.0.0.0/24

(Optional additional subnet) Subnet Name: subnet-asia-south1 Region: asia-south1 IP Range: 10.1.0.0/24

Step 6: Firewall Rules

Do NOT enable default allow-all rules. Create firewall rules separately.

Click CREATE.

CREATE VPC USING CLI (Professional Method)

Step 1: Create Custom VPC

```
gcloud compute networks create my-custom-vpc --subnet-mode=custom
```

Step 2: Create First Subnet

```
gcloud compute networks subnets create subnet-us-central1
--network=my-custom-vpc --region=us-central1 --range=10.0.0.0/24
```

Step 3: Create Second Subnet

```
gcloud compute networks subnets create subnet-asia-south1
--network=my-custom-vpc --region=asia-south1 --range=10.1.0.0/24
```

Step 4: Verify Networks

```
gcloud compute networks list
```

Step 5: Verify Subnets

```
gcloud compute networks subnets list
```

```
Count total subnets: gcloud compute networks subnets list
--format="value(name)" | wc -l
```

```
Count only custom VPC subnets: gcloud compute networks subnets list
--filter="network:my-custom-vpc" --format="value(name)" | wc -l
```

PRACTICES FOR VPC DESIGN

1. Use Custom Mode VPC
 - Full control over IP ranges
 - Avoid automatic subnet sprawl
2. Use RFC1918 Private IP Ranges
 - 10.0.0.0/8
 - 172.16.0.0/12
 - 192.168.0.0/16
3. Plan CIDR Allocation Properly
 - Avoid overlapping ranges
 - Leave room for future expansion
4. Use Regional Subnet Strategy
 - Separate subnets for production, staging, dev
5. Apply Firewall with Least Privilege
 - Do NOT allow 0.0.0.0/0 for SSH/RDP in production
 - Restrict SSH to your public IP (/32)
6. Use Network Tags
 - Apply firewall rules to tagged instances
7. Monitor Subnet Utilization
 - Track IP usage regularly

ARCHITECTURE FLOW

Project | |– VPC (Global) | |– Subnet (Regional) | |– VM Instances |–
Firewall Rules |– Routes

✓ Created Custom VPC manually via console ✓ Created VPC via CLI ✓
Created multiple regional subnets ✓ Verified using gcloud commands ✓
Learned production best practices

End of Document